

Simulating the Chelate Effect

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Supporting Information

ABSTRACT: Despite the rich history of experimental studies focusing on the thermochemistry and kinetics associated with the chelate effect, molecular-level computational studies on the chelate ring opening/closure are scarce. The challenge lies in an accurate description of both the metal ion and its aqueous environment. Herein, we demonstrate that an optimized 12-6-4 Lennard-Jones (LJ) model can capture the thermodynamics and provide detailed structural and mechanistic insights into the formation of ethylenediamine (en) complexes with metal ions. The water molecules in the first solvation shell of the metal ion are found to facilitate the chelate ring formation. The optimized parameters further simulate the formation of bis and tris(en) complexes representing the wide applicability of the model to simulate coordination chemistry and self-assembly processes.

The concept of the chelate ring is a natural outgrowth of Werner's coordination theory of 1893.¹ However, even Werner took eight more years to explicitly assign a cyclic structure to metal ion complexes of acetylacetone.^{2,3} Later, the initial observation of the preferential association of bidentate ligands relative to monodentate ligands to metal ions was termed the "chelate effect" derived from the Greek word *chela*, referring to the great claw of lobsters.⁴ The thermodynamics of these ring systems was quantified in the seminal work of Schwarzenbach in 1952.⁵ The stability constant of ethylenediamine (en) complexes was found to be 10 orders of magnitude higher than that of the corresponding ammonia complexes. Over the past century, inorganic chemists have harnessed the chelate effect for controlled ligand design in various fields ranging from pharmacology⁶ to material science (e.g., MOFs).^{7,8} However, the mechanistic details of this widely employed reaction remain fully to be elucidated.

A quick perusal of the literature spanning over 60 years will reveal, in contrast to the exhaustive list of experimental studies, computational studies of the chelate effect are rare. With both thermodynamic and kinetic aspects of this macrocyclic effect depending on the surrounding solvent molecules, it is essential to include explicit solvent molecules in mechanistic studies.^{9,10} The detailed *ab initio* study by Vallet et al. demonstrate the importance of microsolvated clusters in order to maintain the particle number, thus avoiding spuriously large contributions from translational entropies.¹¹ However, the effectiveness of using a few explicit water molecules with a continuum solvation model is often debated.^{12,13} Nonetheless, when

explicit water molecules are subjected to *ab initio* based molecular dynamics (MD) using, for example, Car–Parrinello MD (CPMD), the steep scaling of these methods restricts them to shorter time scales as seen in the study of Buhl et al. on the binding modes of oxalate to UO₂(oxalate) that can result in poor convergence.¹⁴

Beyond quantum based approaches classical modeling can afford an effective strategy, but the accuracy of these methods using the traditional Lennard-Jones potential (so-called 12-6 potential) has not been effective due to inaccuracies of the model in replicating the thermodynamics of solvation and structural details (e.g., ion-oxygen distances). Figure 1 shows a

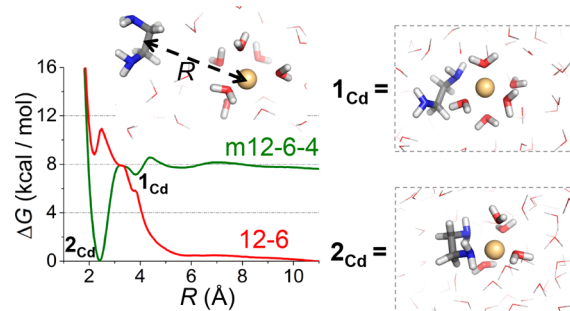


Figure 1. (a) Comparison of PMF profiles for 12-6 (red) and m12-6-4 (green) Cd²⁺ ion parameters interacting with ethylenediamine.

conventional MD calculation using the 12-6 potential simulating the formation of the chelate complex to be a thermodynamically less stable state than the separated species. Alternatively, we can employ our newly developed 12-6-4 LJ nonbonded model that can simultaneously reproduce thermodynamic and structural properties of broad range of aqueous ions.^{15,16} The model includes a r^{-4} term tuned to incorporate the ion-induced dipole interactions (Figure 1, green). Prior studies demonstrated this term reproduced the induction contribution but was never incorporated in a generalized model to simulate the reactivity of metal ions with diverse ligands.^{17,18} Important contributions from many-body interactions toward the solvation energy of metal ions further preclude the use of conventional 12-6 LJ nonbonded model.^{19,20} While sophisticated force fields like SIBFA (sum of interactions between fragments *ab initio* computed),²¹ NEMO (nonempirical molecular orbital),²² and CTPOL include local polarization effects and charge transfer,²³ the lengthy para-

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metrization and higher computational cost of the methods restrict their applicability. However, Panteva et al. found the 12-6-4 model to be a more accurate model over 12-6 models in simulating the interactions between the Mg^{2+} ion and nucleotide bases in aqueous solution.²⁴ Subsequent application of the model provided an improved understanding of metal ion transfer in the hammerhead ribozyme catalytic reaction.²⁵ These initial studies provide a strong rationale toward the applicability of the 12-6-4 model to study solvent effects in metal complexation reactions.

Herein, we apply our 12-6-4 LJ nonbonded model for metal ions along with TIP3P water model to study the ion complexation of Zn^{2+} , Ni^{2+} , Cu^{2+} and Cd^{2+} with en.²⁶ MD calculations over 8 ns were performed along the reaction coordinate between the metal ion and the center of mass of the ligand to study chelate ring formation. We present the reaction between Cd^{2+} and en as a representative example for the study of this process. The conventional 12-6 LJ model (Figure 1, red) suggests the chelate complex to be thermodynamically unfavorable relative to the solvated metal ion by 8.8 kcal/mol. The default 12-6-4 LJ model provides similar results with a small additional stability for the complex owing to some of the charge-induced dipole interaction being picked up by the default parameters (see Supporting Information). Furthermore, both 12-6 and the default 12-6-4 depict the presence of only one minima corresponding to the absence of any stable structure involving monocoordination of en with Cd^{2+} . This figure cautions against the use “off the shelf” ion parameters because they will likely be unable to model processes they were not designed to represent. At a minimum, careful validation is called for when using available metal ion models on new problems. Upon optimizing the 12-6-4 LJ (called m12-6-4) parameters between the metal ion and the nitrogen atoms of en to reproduce the experimental binding energies of the $[\text{Cd}(\text{en})(\text{H}_2\text{O})_4]^{2+}$ complex, the obtained potential of mean force (PMF) plot (Figure 1, green) captures two different minima corresponding to the ligand bound to the metal ion in monodentate (1_{Cd}) as well as in a bidentate mode (2_{Cd} , chelate complex).

Figure 2 represents snapshots of different stationary points in the energy profile for Cd^{2+} interacting with en obtained with the m12-6-4 ion model. It should be noted that the highlighted water molecules around the metal ion only represent a snapshot of the structural features around the ion that are in reality constantly undergoing exchange with the bulk water

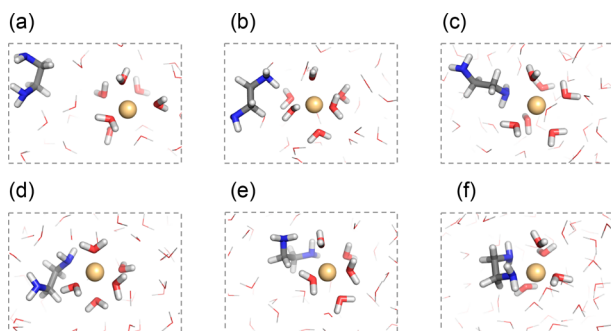


Figure 2. Snapshots at (a) intermolecular distance $R = 8.00$ Å, (b) $R = 4.80$ Å, (c) $R = 4.35$ Å (TS1), (d) $R = 3.80$ Å (1_{Cd}), (e) $R = 3.23$ Å (TS2), and (f) 2.40 Å (2_{Cd}) in the formation of $[\text{Cd}(\text{en})(\text{H}_2\text{O})_4]^{2+}$. R is the intermolecular distance between the center of mass of the ligand and ion.

molecules. In Figure 2, we start from the solvated metal ion (at 8 Å, Figure 2a) and describe the reaction process associated with the formation of the chelate ring complex (at 2.4 Å, Figure 2f). The en ligand was parametrized to access both the *gauche* and *anti* conformers (for further details, see the SI and Figure S1). The free ligand itself can act as a hydrogen bond donor or acceptor in interactions with water molecules, while the Cd^{2+} ion is bound to six water molecules in an octahedral geometry.

The association of the metal ligand proceeds from the first minima ($\Delta G_{\text{min1}} = 7.8 \pm 0.3$ kcal/mol relative to the chelated complex) corresponding to the fully solvated hexacoordinated metal ion and unbound ligand at $R = 4.80$ Å (see Figure 2b). The minimum corresponds to the solvent separated ion ligand pair, wherein en engages in intermolecular hydrogen bonding with a water molecule present in the first solvation shell of Cd^{2+} . As the ligand approaches closer to the metal ion, at $R = 4.35$ Å we have transition state TS1 ($\Delta G_1^\ddagger = 8.4 \pm 0.1$ kcal/mol) corresponding to the state where the ligand starts associating with the metal ion while concomitantly one of the water molecules begins to depart from the ion coordination sphere. Note that the small barrier height of 0.6 kcal/mol relative to the separated ion ligand pair is similar (1.1 kcal/mol) to the one found in formation of $[\text{Cd}(\text{MeNH}_2)(\text{H}_2\text{O})_5]^{2+}$ when the pairwise parameter was optimized to reproduce the experimental binding energy of the MeNH_2 complex (Figure S9 in Supporting Information). We find the coordination number of the metal ion increases from 6 to 7 in these transient states (see Figure 2c) corresponding to an associative mechanism. The average bond lengths $\text{Cd}-\text{O}_{\text{wat}}$ and $\text{Cd}-\text{N}$ distances for the outgoing water and the incoming amine are found to be 2.40 and 2.61 Å respectively. The reaction then proceeds to form 1_{Cd} at $R = 3.80$ Å (see Figure 2d) corresponding to the ligand in the *anti* conformation coordinating with Cd^{2+} in a monodentate fashion. The average $\text{Cd}-\text{N}$ distances were found to be 2.36 and 5.18 Å for the bound and unbound nitrogen atoms.

A further decrease in the intermolecular distance ($R < 3.50$ Å) results in the ligand losing its conformational flexibility and being present more in the *gauche* conformer allowing both of the nitrogen atoms to interact with the ion. At $R = 3.23$ Å (see Figure 2e), we obtain transition state TS2 ($\Delta G_2^\ddagger = 7.9$ kcal/mol) corresponding to a ligand exchange between the second amine end of en and a water molecule. The average $\text{Cd}-\text{O}_{\text{wat}}$ and $\text{Cd}-\text{N}$ distances for the outgoing water and the incoming nitrogen atom are found to be 2.40 and 3.96 Å, respectively. We observe that while the lone pair of incoming nitrogen still faces the metal ion, the water molecules closer to this nitrogen end reorient themselves to hydrogen bond with this nitrogen atom and interact with the metal ion simultaneously. This bridging interaction likely reduces the free energy barrier for the formation of the second $\text{Cd}-\text{N}$ bond. The gradual decrease in intermolecular distance to $R = 2.40$ Å results in chelate ring closure forming the final complex. Among the different metal ions, we find the highest barrier heights for Ni en complexes corresponding to the stronger Ni–N bonds. The reaction profile of metal ligand association is documented in a video given in the Supporting Information.

A 12-6-4 parameter set was then specifically built to reproduce the experimental binding energy of methylamine complexes of different metal ions. These parameters were then used to calculate the free energies of formation for bis(methylamine) and compare against prior energies

of en complex of Cd^{2+} (see Figure 3). The global minimum as expected for the en is deeper than the one for bis-

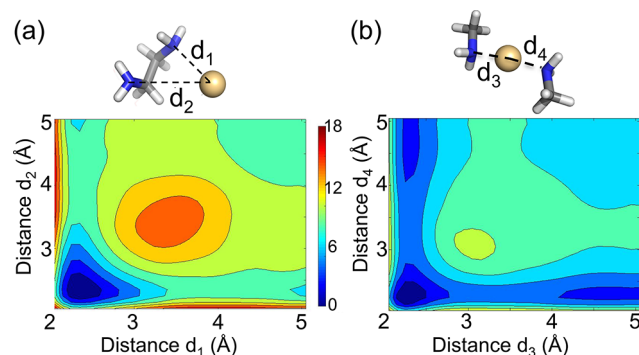


Figure 3. Comparison of 2D PMF contour plots toward formation of (a) en and (b) bis(methylamine) complexes of Cd^{2+} .

(methylamine) where the free energy difference corresponds to the chelate effect. Further, the constrained 2D PMF contour plots highlights the involvement of a rotational movement of the en for ring closure relative to the sole translational motion associated with the formation of $\text{Cd}(\text{II})$ complex of bis-(methylamine). *Ab initio* DLPNO-CCSD(T) calculations on model systems considering first solvation shell overestimate the chelate effect by 8 kcal/mol. Additional calculations with the pairwise parameters for Cd en interactions reproduced the experimental binding energies of bis(en) and tris(en) complexes of $\text{Cd}(\text{II})$ highlighting the transferability of the parameters in a different environment. The excellent agreement with experimental reaction energies is partly due to cancellation of errors in the incomplete representation of charge-transfer and many-body effects across the reactants and products. We recommend the present model for simulating metal ligand complexes with structural features (bond distances and coordination numbers) similar to that of the solvated metal ion.

The present parameter sets allowed us to perform real time simulations on 10 Cd^{2+} and 30 en ligand molecules in a box of water to form mono-, bis-, and tris(en) complexes of Cd^{2+} (Figure 4). We find that a 0.1 M system is saturated by the

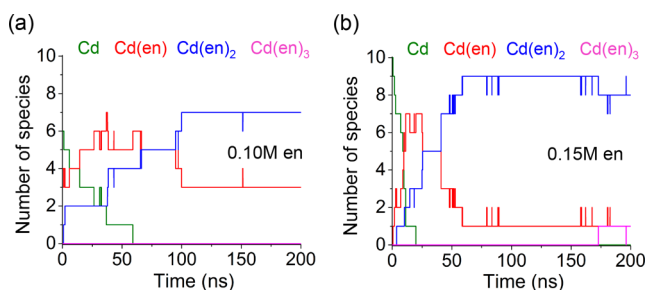


Figure 4. Variation of number of molecules of different species in real time simulation for (a) 0.10 M and (b) 0.15 M ethylenediamine (en) solutions.

bis(en) Cd^{2+} after 100 ns of simulation, while higher concentrations (0.15 M) are required to observe the formation of tris(en) Cd^{2+} in a similar time-scale. We note that the present calculations are obtained from a single trajectory for each concentration but that an ensemble of such trajectories would afford a better comparison with experiment.

Overall, our model effectively simulates the chelate effect and provides molecular-level mechanistic details of the solvent assisted chelate ring opening/closure exchange reaction. The hydrogen bonding interactions between water molecules near Cd^{2+} and en facilitates the ring closure. Finally, the present model can be readily applied to studies of related metal ion complexation processes in condensed phases.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/jacs.8b09371.

Details of the simulation protocols, parameter fitting, computed binding free energies for different divalent ions and en for m12-6-4 ion model (PDF)

Video showing reaction profile of metal ligand association (AVI)

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Notes

The authors declare no competing financial interest.

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(26) For complete details of the computational protocols, see Section S1 in Supporting Information.

■ NOTE ADDED AFTER ASAP PUBLICATION

Due to a production error, an incorrect version of the Supporting Information was published on November 6, 2018. The correct version was reposted on November 14, 2018.